

Resuelva la desigualdad

$$9(x-21)^2 < -2 \cdot |63-3x| + 3$$

$$\bullet |63-3x| = \begin{cases} 63-3x & \text{si } 63-3x \geq 0 \\ & 63 \geq 3x \\ & 21 \geq x \\ -(63-3x) & \text{si } 63-3x < 0 \\ & 63 < 3x \\ & 21 < x \end{cases}$$

^{no} $\bullet 9(x^2-21)^2 < -2 \cdot (63-3x) + 3 \quad \text{si } 21 \geq x$

$$9(x^2 - 2 \cdot x \cdot 21 + 21^2) < -126 + 6x + 3$$

$$9(x^2 - 42x + 441) < 6x - 123$$

$$9x^2 - 378x + 3969 < 6x - 123$$

$$9x^2 - 378x - 6x + 3969 + 123 < 0$$

$$9x^2 - 384x + 4092 < 0$$

$$3 \cdot (3x^2 - 128x + 1364) < 0$$

Caso + -

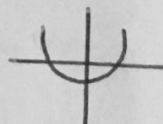
$$3 > 0 \quad \wedge \quad 3x^2 - 128x + 1364 < 0$$

VERDADERO

$\boxed{\mathbb{R}}$

$$\begin{cases} x_1 = 22 \\ x_2 = 62/3 \end{cases}$$

$\boxed{(62/3; 22)}$



$\mathbb{R} \wedge (62/3; 22)$

$\boxed{(62/3; 22)}$

Caso - +

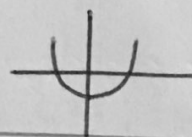
$$3 < 0 \quad \wedge \quad 3x^2 - 128x + 1364 > 0$$

FALSO

$\boxed{\emptyset}$

$$\begin{cases} x_1 = 22 \\ x_2 = 62/3 \end{cases}$$

$\boxed{(-\infty; 62/3) \cup (22; +\infty)}$



$\emptyset \wedge (-\infty; 62/3) \cup (22; +\infty)$

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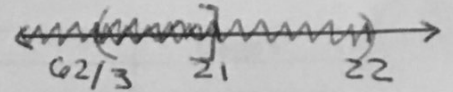
①

$\boxed{\emptyset}$

$$\text{Sol } ① = \{(62/3; 22) \cup \emptyset\} \wedge (-\infty; 21]$$

$$\boxed{\text{Sol } ① = (62/3; 21]}$$

B



$$\cdot 9(x^2 - 21)^2 < -2 \cdot (-63 + 3x) + 3 \quad \text{si} \quad 21 < x$$

$$9(x^2 - 2 \cdot x \cdot 21 + 21^2) < 126 - 6x + 3$$

$$9(x^2 - 42x + 441) < -6x + 129$$

$$9x^2 - 378x + 3969 < -6x + 129$$

$$9x^2 - 378x + 6x + 3969 - 129 < 0$$

$$9x^2 - 372x + 3840 < 0$$

$$3(3x^2 - 124x + 1280) < 0$$

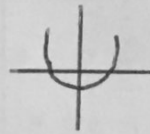
Caso + -

$$3 > 0 \quad \wedge \quad 3x^2 - 124x + 1280 < 0$$

VERDADERO

$\boxed{\mathbb{R}}$

$$\begin{cases} \rightarrow x_1 = 64/3 \\ \rightarrow x_2 = 20 \end{cases}$$



$$\boxed{(20; 64/3)}$$

$$\mathbb{R} \wedge (20; 64/3)$$

$$\boxed{(20; 64/3)}$$

Caso - +

$$3 < 0 \quad \wedge \quad 3x^2 - 124x + 1280 < 0$$

FALSO

$\boxed{\emptyset}$

$$\begin{cases} \rightarrow x_1 = 64/3 \\ \rightarrow x_2 = 20 \end{cases}$$

$$\boxed{(-\infty; 20) \cup (64/3; +\infty)}$$

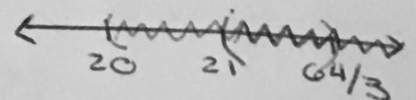
$$\emptyset \wedge (-\infty; 20) \cup (64/3; +\infty)$$

$\boxed{\emptyset}$

$$\text{Sol } ② = \{(20; 64/3) \cup (\emptyset)\} \wedge (21; +\infty)$$

$$\boxed{\text{Sol } ② = (21; 64/3)}$$

B



②

$$\text{Sol final} = \text{sol } ① \cup \text{sol } ②$$

$$\text{Sol final} = (62/3, 21] \cup (21, 64/3)$$

$$\text{Sol final} = (62/3, 64/3) \quad \text{B}$$

Sea f una función con dominio $(0; +\infty)$. Se sabe que las soluciones de las siguientes inequaciones son las indicadas en cada caso:

• $f(x) > x - 25$ tiene como solución el conjunto $(25; +\infty)$

• $f(x) < x - 25$ tiene como solución el conjunto $(23; 25)$

Resolver la inequación

$$\frac{f(x) + 1}{x - 24} < 1$$

$$\frac{f(x) + 1}{x - 24} - 1 < 0$$

$$\frac{f(x) + 1 - x + 24}{x - 24} < 0$$

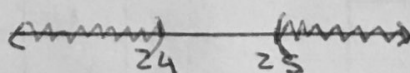
$$\frac{f(x) - x + 25}{x - 24} < 0$$

caso + -

$$f(x) - x + 25 > 0 \quad \wedge \quad x - 24 < 0$$

$$f(x) > x - 25 \quad \wedge \quad x < 24$$

$$(25; +\infty) \quad \wedge \quad (-\infty; 24)$$



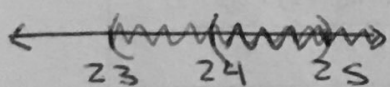
$\emptyset$ B

caso - +

$$f(x) - x + 25 < 0 \quad \wedge \quad x - 24 > 0$$

$$f(x) < x - 25 \quad \wedge \quad x > 24$$

$$(23; 25) \quad \wedge \quad (24; +\infty)$$

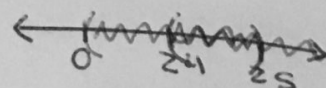


$(24; 25)$ B

$$\text{sol} = \{\emptyset \cup (24; 25)\} \wedge \text{Dom}(f)$$

$$\text{sol} = (24; 25) \quad \text{B}$$

③



Sea f una función con dominio el intervalo $(10; +\infty)$ y que tiene al eje x como asíntota horizontal a derecha.

calcular el siguiente límite:

$$\lim_{x \rightarrow +\infty} \left(\frac{\sin(13 \cdot f(x))}{f(x)} + f(x) \cdot \sin(x \cdot f(x)) \right)$$

$$\text{Dom}(f) = (10; +\infty)$$

ejemplo $f(x) = \frac{-1}{\sqrt{x-10}}$

$$\lim_{x \rightarrow +\infty} f(x) = 0$$

$$\lim_{x \rightarrow +\infty} \left(\underbrace{\frac{\sin(13 \cdot \overset{0}{f(x)})}{\underset{0}{f(x)}}}_{\text{indeterminación "0/0"}} + \underset{0}{f(x)} \cdot \underbrace{\sin(\overset{+\infty}{x} \cdot \overset{0}{f(x)})}_{\text{ACOTADO}} \right)$$

$$\lim_{x \rightarrow +\infty} \left(\left(\frac{\sin(13 \cdot f(x))}{f(x)} + f(x) \cdot \sin(x \cdot f(x)) \right) \cdot \frac{13}{13} \right)$$

$$\lim_{x \rightarrow +\infty} \left(\frac{\sin(13 \cdot f(x))}{f(x)} \cdot \frac{13}{13} + f(x) \cdot \sin(x \cdot f(x)) \cdot \frac{13}{13} \right)$$

$$\lim_{x \rightarrow +\infty} \left(\underbrace{\frac{\sin(13 \cdot f(x))}{13 \cdot f(x)}}_1 \cdot \underbrace{13}_{13} + \underbrace{\frac{f(x) \cdot \sin(x \cdot f(x))}{0}}_{\text{ACOTADO}} \cdot \underbrace{\frac{13}{13}}_1 \right)$$

$$\lim_{x \rightarrow +\infty} 1 \cdot 13 + 0 \cdot 1$$

Propiedades:

$$\lim_{x \rightarrow +\infty} 13$$

B

Si $\lim_{x \rightarrow x_0} f(x) = 0 \rightarrow \lim_{x \rightarrow x_0} \frac{\sin(f(x))}{f(x)} = 1$

$0 \cdot \text{ACOTADO} = 0$

$-1 \leq \sin(x) \leq 1$ ACOTADA

Sea $(a_n)_{n \in \mathbb{N}}$ una sucesión de términos positivos. Se sabe que

$$\lim_{n \rightarrow \infty} \frac{a_{n+1}}{a_n} = 12$$

(a) calcule el límite de la sucesión $(a_n)_{n \in \mathbb{N}}$

(b) calcule el siguiente límite: $\lim_{n \rightarrow \infty} \frac{(11)^n}{a_n}$

(a) Usamos el criterio de D'Alembert

$$\lim_{n \rightarrow \infty} \frac{a_{n+1}}{a_n} = 12 > 1$$

entonces

$$\boxed{\lim_{n \rightarrow \infty} a_n = +\infty} \quad B$$

$$\lim_{n \rightarrow \infty} \frac{a_{n+1}}{a_n}$$

$\left\{ \begin{array}{l} < 1 \text{ entonces } \lim_{n \rightarrow \infty} a_n = 0 \\ > 1 \text{ entonces } \lim_{n \rightarrow \infty} a_n = +\infty \\ = 1 \text{ criterio no decide} \end{array} \right.$

(b) Usamos el criterio de D'Alembert

$$a_n = \frac{(11)^n}{b_n} \quad a_{n+1} = \frac{(11)^{n+1}}{b_{n+1}}$$

$$\lim_{n \rightarrow \infty} \frac{a_{n+1}}{a_n}$$

$$\lim_{n \rightarrow \infty} 11 \cdot \frac{1/12}{1/20}$$

$$\lim_{n \rightarrow \infty} \frac{(11)^{n+1}}{a_{n+1}} : \frac{(11)^n}{a_n}$$

$$\lim_{n \rightarrow \infty} 11/20 < 1$$

entonces

$$\lim_{n \rightarrow \infty} \frac{(11)^{n+1}}{a_{n+1}} \cdot \frac{a_n}{(11)^n}$$

$$\boxed{\lim_{n \rightarrow \infty} \frac{(11)^n}{a_n} = 0} \quad B$$

$$\lim_{n \rightarrow \infty} \frac{11 \cdot 11^n}{11^n} \cdot \frac{a_n}{a_{n+1}}$$

$$\lim_{n \rightarrow \infty} 11 \cdot \frac{a_n}{a_{n+1}} \cdot \frac{1/20}{1/12} \quad ??$$

Propiedades:

$$a^{r+s} = a^r \cdot a^s$$

$a > 0 ; r, s \in \mathbb{Q}$

(5)